

October 1, 2021

Supersource (pseudocode)

Supersource (G)

$n := |V|$, $i := 1$, $j := n$

while $i \neq j$ do

if $\text{Adj}[i, j] = 0$

then $i := i + 1$

else $j := j - 1$

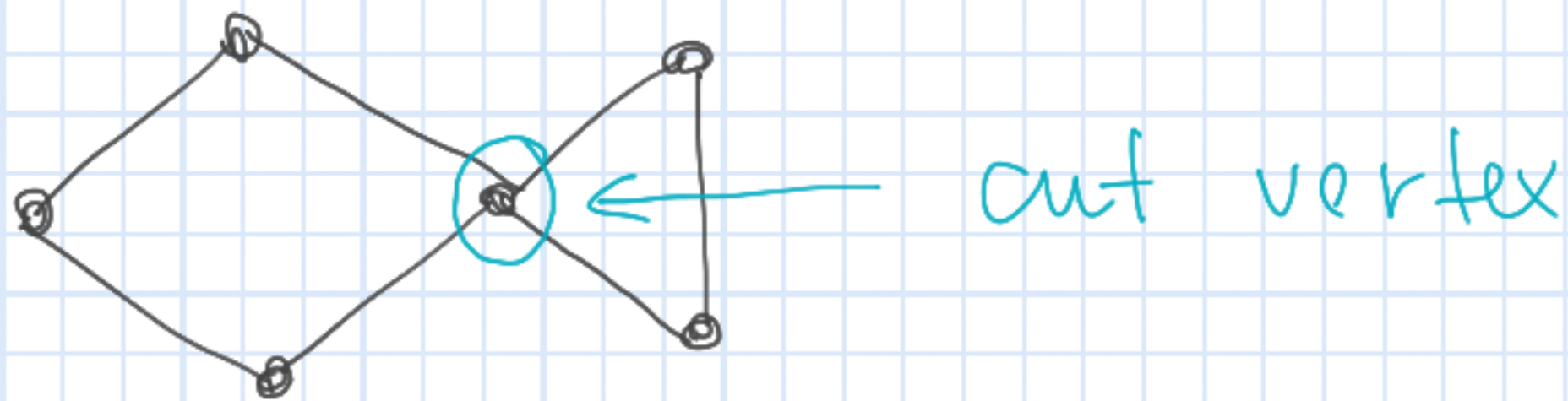
1st phase $\rightarrow i$ remains

2nd phase {
: for $k = 1$ to n do
 if $k \neq i$ AND ($\text{Adj}[i, k] \neq 1$ OR $\text{Adj}[k, i] \neq 0$)
 then return nil /* i is not a S.S.
return i /* i is a S.S.

Traversals of Graphs

The aim is to explore the structure of a graph

- Is the graph connected?
- Is the graph bipartite?
- Does the graph contain a cycle?
- Is it possible to remove one vertex to disconnect the graph?



The two most popular traversal methods

→ breadth first search (BFS)

→ depth first search (DFS)

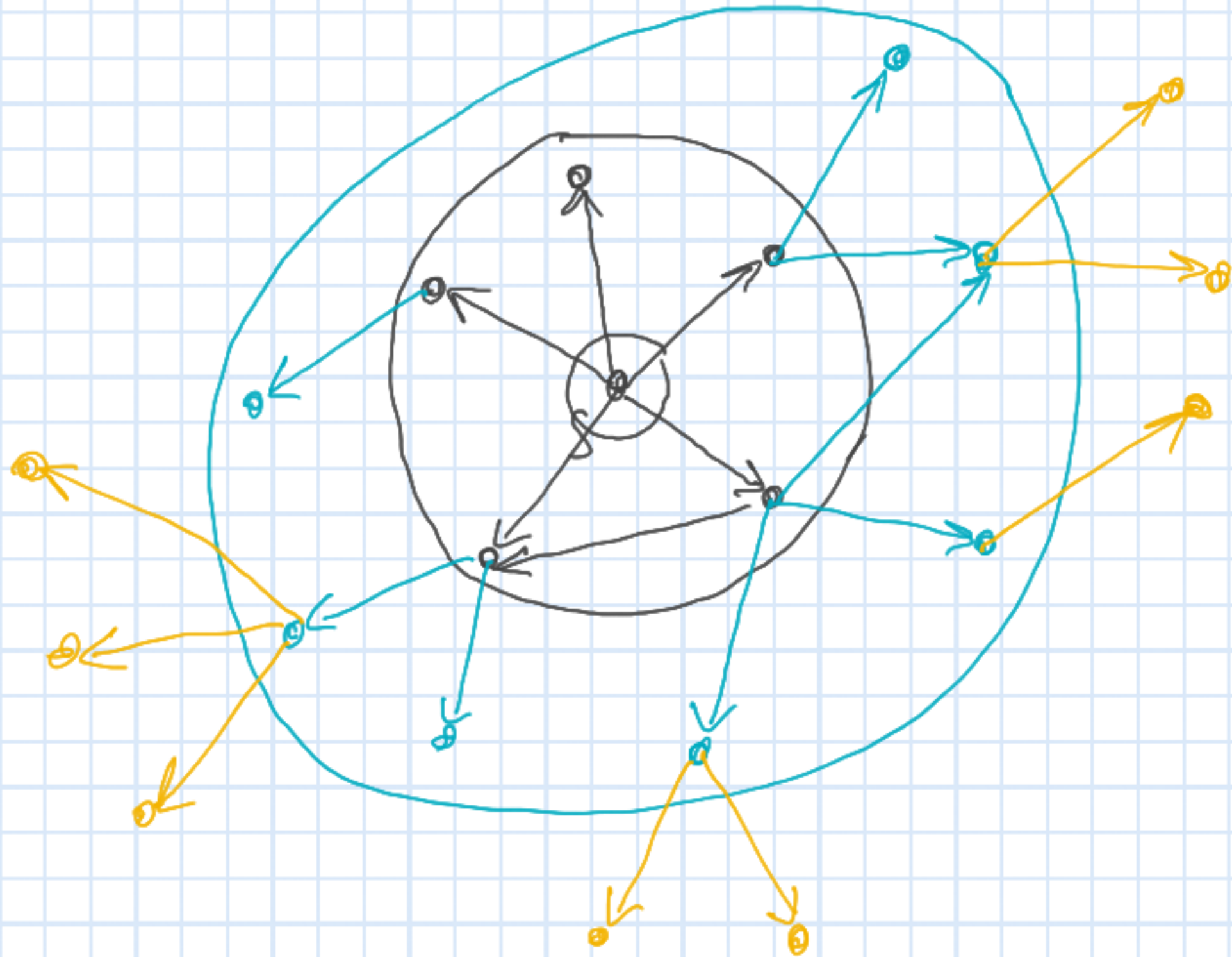
BFS

We are given a directed or undirected graph $G = (V, E)$ by its adjacency lists, and a source vertex $s \in V$

We explore G from s and we discover the part of G that can be reached from s along edges of G (this is not necessarily the whole graph).

First we discover the (first) neighbour of s , then the neighbours of the first neighbours (not discovered yet), and so on.

Visualization



"the boundary between the discovered and undiscovered parts is extended uniformly"

BFS returns all vertices that can be reached from s WITH the shortest paths from s to these vertices $\rightarrow$ BFS tree

Organisation

- colouring the vertices

white ↗	grey ↑	black ↖
undiscovered	discovered but it may have undiscovered neighbours	discovered as well as all of its neighbours

- a data structure queue is used to store the grey vertices

BFS (G, s)

for each $u \in V \setminus \{s\}$

$\text{colour}[u] := \text{white}$

$d[u] := \infty$ $\leftarrow$ distance of u from s

$\text{parent}[u] := \text{nil}$ $\leftarrow$ parent of u in the BFS tree

$\text{colour}[s] := \text{grey}$

$d[s] := 0$

$\text{parents}[s] := \text{nil}$

: initialize $Q \leftarrow \text{queue}$

Enqueue(Q, s) $\leftarrow$ insert into Q

while Q is not empty do

$u := \text{Dequeue}(Q) \leftarrow$ delete from Q

 for each $v \in \text{Adj}[u]$ do $\leftarrow$ Adj list of u

 if colour $[v] = \text{white}$ then

 colour $[v] := \text{grey}$

 parent $[v] := u$

$d[v] := 1 + d[u]$

 Enqueue(Q, v)

! colour $[u] := \text{black}$

Time complexity

$\Theta(|V| + |E|)$ linear time algorithm

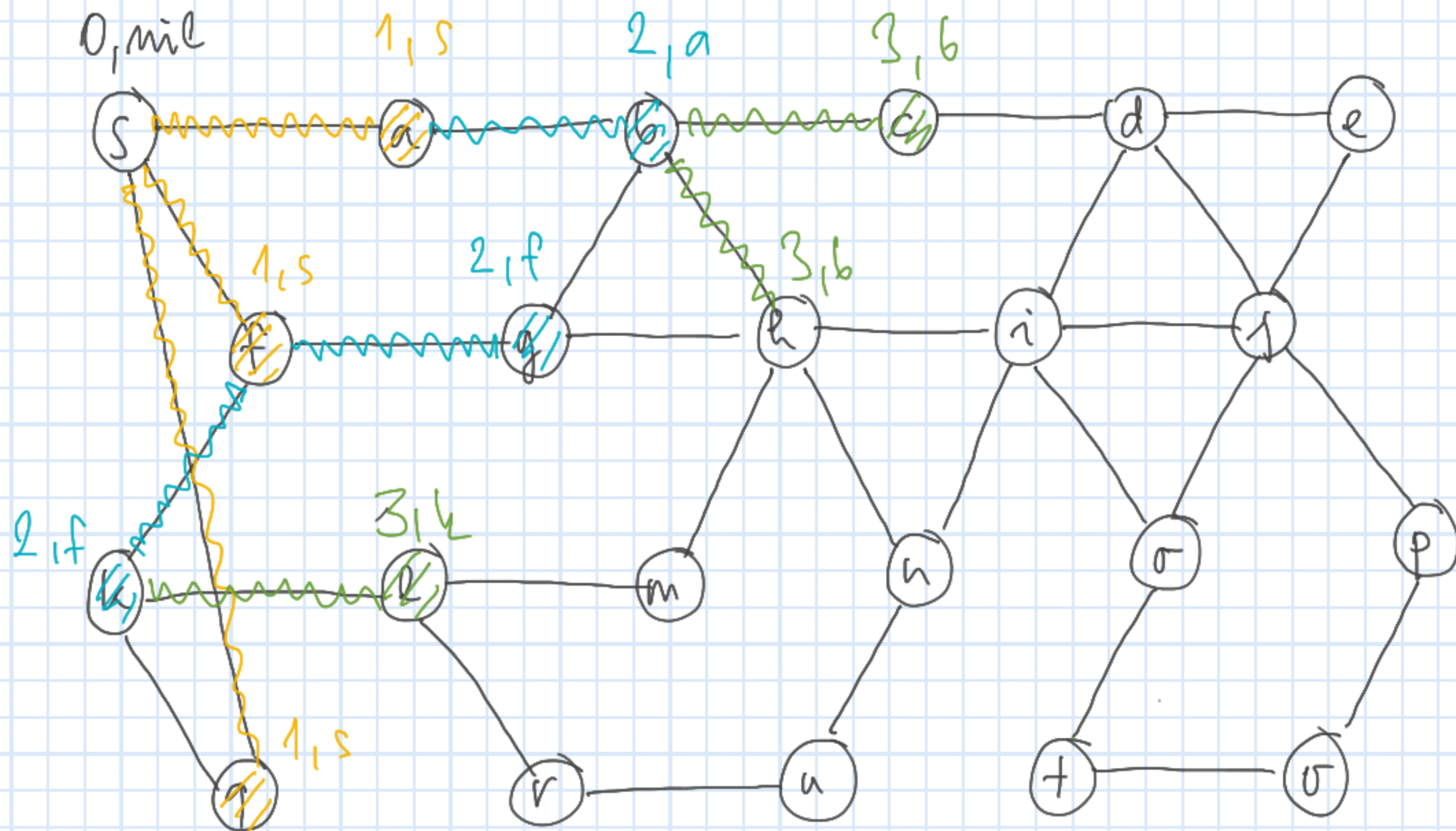
The cost of the initialization is $\Theta(|V|)$.

We enqueue and dequeue each vertex at most once, and an adjacency list is read once when we dequeue the corresponding vertex.

Correctness

Less clear, we don't prove it.

Example



Queue

S

a	f	g
---	---	---

f	a	b
---	---	---

g	b	g	k
---	---	---	---

6/g/k

g	h	c	h
---	---	---	---

$$\boxed{h | c | h}.$$

c | h | r

HW to finish

